

bom — sheet: BOM

Component	Mass g/cell	Mass kg/kWh	Basis / source
Positive electrode coating (LNMO)	22.72	0.894	cell/r8_d6_energy_dfn.json:layer_kg_m2[Positive electrode coating (LNMO)] x area_m2
Negative electrode coating (graphite)	13.04	0.513	cell/r8_d6_energy_dfn.json:layer_kg_m2[Negative electrode coating (graphite)] x area_m2
Positive current collector (Al)	4.44	0.175	cell/r8_d6_energy_dfn.json:layer_kg_m2[Positive current collector (Al)] x area_m2
Negative current collector (Cu)	11.04	0.435	cell/r8_d6_energy_dfn.json:layer_kg_m2[Negative current collector (Cu)] x area_m2
Separator	0.26	0.01	cell/r8_d6_energy_dfn.json:layer_kg_m2[Separator] x area_m2
Electrolyte (EC/EMC + LiPF6)	6.27	0.247	design estimate: pore volume 5.804 cm3 x 1.2 g/cm3 (literature) x 0.9 fill
Conductive additive (positive)	estimate 0.34 (1.5 wt% of 22.72 g)	—	literature default 1.5 wt% (estimate, not parameterized)
Binder (positive)	estimate 0.45 (2.0 wt% of 22.72 g)	—	literature default 2.0 wt% (estimate, not parameterized)
Conductive additive (negative)	estimate 0.13 (1.0 wt% of 13.04 g)	—	literature default 1.0 wt% (estimate, not parameterized)
Binder (negative)	estimate 0.39 (3.0 wt% of 13.04 g)	—	literature default 3.0 wt% (estimate, not parameterized)
Enclosure and tabs	Not modeled	—	beyond pure simulation boundary
TOTAL (electrolyte excluded, contract caliber)	51.5	2.027	cell/r8_d6_energy_dfn.json:mass_kg
TOTAL incl. electrolyte estimate	57.77	2.274	computed
Total energy	25.41	Wh	cell/r8_d6_energy_dfn.json:energy_wh